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NITI Aayog



SECTOR AT A GLANCE 04

Higher Education in India

DRAFT

SECTOR AT A GLANCE 04

HIGHER EDUCATION IN INDIA

EDUCATION IN INDIA

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Over the past two decades, the Indian higher education system has experienced remarkable growth, with a 4.5 times in the number of universities and colleges.¹ Enrolment in higher education institutions (HEIs) has surged almost fourfold, from around 9 million in 2001 to 43 million in 2021-22, supported by a Gross Enrolment Ratio (GER) of 28.4, surpassing comparable lower and middle-income countries.² A series of policy shifts over past decades, including support for greater private sector participation as well as increasing devolution of authority as well as budgets to state governments, have enabled the rising capacity of India's higher education system.³

Emerging opportunities in the higher education landscape in India include promoting multidisciplinary learning, advancing faculty professional development, strengthening research capabilities, ensuring inclusivity, integrating digital learning, and advancing internationalisation. Key issues pertaining to higher education include limited industry-relevant skills training and specialised vocational offerings, shortages of trained and motivated teachers, a fragmented system of HEIs, an insufficient number of qualified researchers and quality research output, and financing and challenges that prevent critical investments.

Given the challenges and opportunities, key government priorities include curriculum and pedagogy, faculty recruitment, training, and development, research, innovation, and entrepreneurship, equity and inclusion, vocational education within higher education, resourcing, governance, and assessment of higher education and financing. Ongoing central government policies including, but not limited to the National Education Policy (NEP), the National Higher Education Qualification Framework, and the National Credit Framework are aimed at re-energising the Higher Education system to deliver high-quality education, with equity and inclusion. These efforts, along with several state- and local-level policies and initiatives, seek to ensure that India's higher education system develops thoughtful, well-rounded, and creative graduates.

Going forward, the NEP has outlined a wide range of reforms to develop a high-quality, well-governed, and accessible higher education system. These focus on developing institutional structures and regulations that enable effective accreditation at scale, the publishing and financing of high-quality research, accessible and student-centric curricula, competent and well-motivated faculty, and a supportive environment for international campuses and student mobility will be critical to meet the vision of developing India into a global knowledge superpower. This note presents key trends related to different components of higher education in India and some ideas for the way forward.

Figure 1: Key higher education statistics

4.5 times

Increase in the number of colleges and universities in India since 2000

28.4%

Gross Enrolment Ratio in HEIs in 2021-22

INR 44,094 crore

Budget allocation for HE department for FY 2023-24

48%

Female students enrolled in HEIs in 2021-22

Source: Ministry of Education, All India Survey of Higher Education, 2021-22; Ministry of Education, Ministry of Education has been allocated INR 1,12,899.47 crore, the highest allocation ever to the Education Ministry – Shri Dharmendra Pradhan, 2023

Note: The Gross Enrolment Ratio measures the number of students enrolled in higher education as a share of the total eligible population in the 18-23 age group.

¹ Ministry of Information and Broadcasting, Growth in the Number of Universities and Colleges, 2022; Ministry of Education, All India Survey of Higher Education, 2020-21

² Ministry of Education, All India Survey of Higher Education, 2021-22; Brookings, Reviving Higher Education in India, 2019

³ Brookings, Reviving Higher Education in India, 2019

Key trends

This section dives deeper into India's higher education landscape, outlining statistics and trends on the following topics. The analysis of these key levers is crucial from a policy planning perspective.

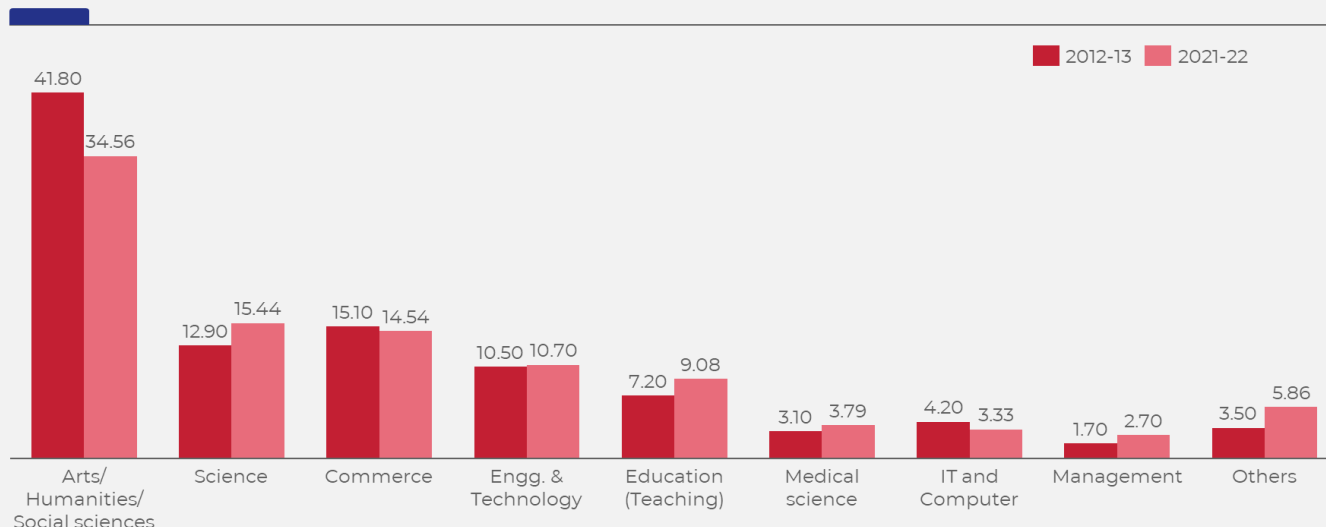
1. Curriculum and pedagogy
2. Faculty Recruitment, Training, and Development
3. Research, innovation, and entrepreneurship
4. Equity and inclusion
5. Vocational education within higher education
6. Resourcing, Governance, and Assessment of higher education
7. Financing for Higher Education

Curriculum and pedagogy

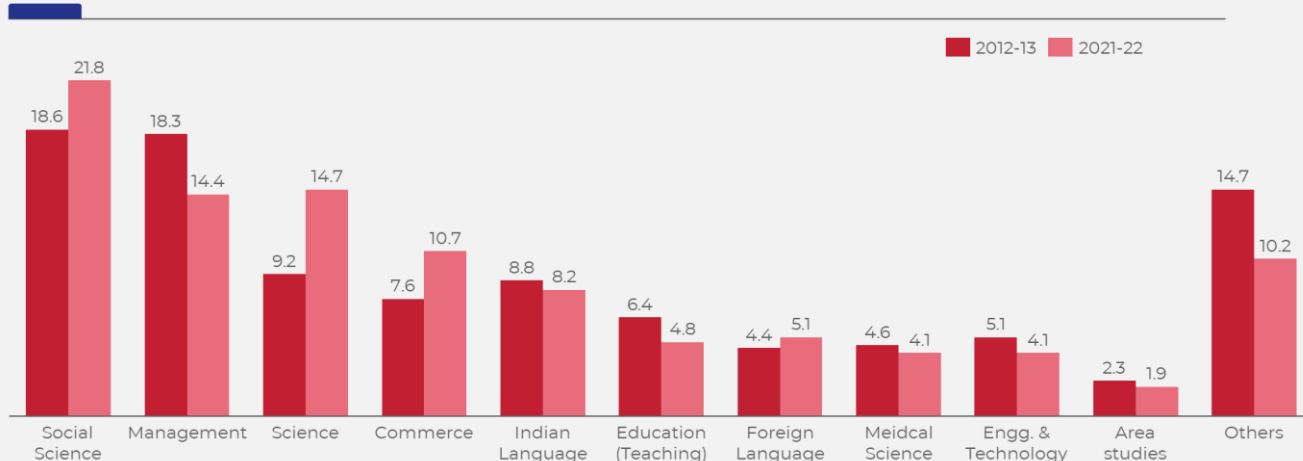
Curriculum and pedagogy in the context of higher education includes fostering high-quality holistic and multidisciplinary education through curriculum flexibility and engaging course options, alongside reformed pedagogy through greater emphasis on communication, discussion, debate, research, and opportunities for cross-disciplinary and interdisciplinary thinking.

Figure 2: Discipline-wise graduations at the UG and PG levels (2012-13 and 2021-22)

Discipline-wise graduations at the UG level, 2021-22, in % of total students



Discipline-wise graduations at the PG level, 2021-22, in % of total students



Source: Ministry of Education, All India Survey of Higher Education 2012-13 and 2021-22

Enhancing curriculum and leveraging experiential learning in HEIs emerge as focus:

According to the UGC, it is crucial to create an interdisciplinary curriculum that emphasises practical application, cultural enrichment, and prepares graduates for employment, thereby facilitating their holistic professional development. Ministry of Education has underscored the importance of contextual and application-oriented pedagogical methods, such as topic-centred clubs and activities initiated by students in collaboration with faculty and relevant experts.⁴ Additionally, there is focus on imparting “industry 4.0” skills like applications of AI in HE courses.

Government plans to increase multi-disciplinary course offerings in HEIs:

The Ministry of Education aims to have least one large multidisciplinary HEI in or near every district by 2030.⁵ The UGC plans to create multi-disciplinary learning environments by incorporating internships, community service, and skill-based courses on subjects such as entrepreneurship in existing course offerings to increase opportunities for graduate-level jobs across government, industry, and academic circles.⁶

Internationalisation of higher education curricula is a key priority:

One of India’s key educational priorities is to ensure that all educational programs, their curriculum, and teaching methods, whether delivered in classrooms, online, or through Open Distance Learning (ODL), will strive to attain international standards of quality. The UGC plans to collaborate with faculty in leading international universities to add a global perspective in Indian curricula, enter into twinning agreements or semester abroad programs that allow Indian students to complete a part of their degree in another country at a partnering institution, and promote faculty exchange programs to increase faculty exposure to foreign curriculum and pedagogies.⁷

Faculty Recruitment, Training, and Development

This includes comprehensive faculty development for the higher education system, including sound recruitment, continuous professional development, positive working environments and service conditions, and provision of autonomy.

Teacher recruitment in India is either directly to higher education institutes or via civil services:

⁴ Ministry of Education, National Education Policy, 2020

⁵ Ministry of Education, National Education Policy, 2020

⁶ Ministry of Education, Year End Review - Department of Higher Education, 2021

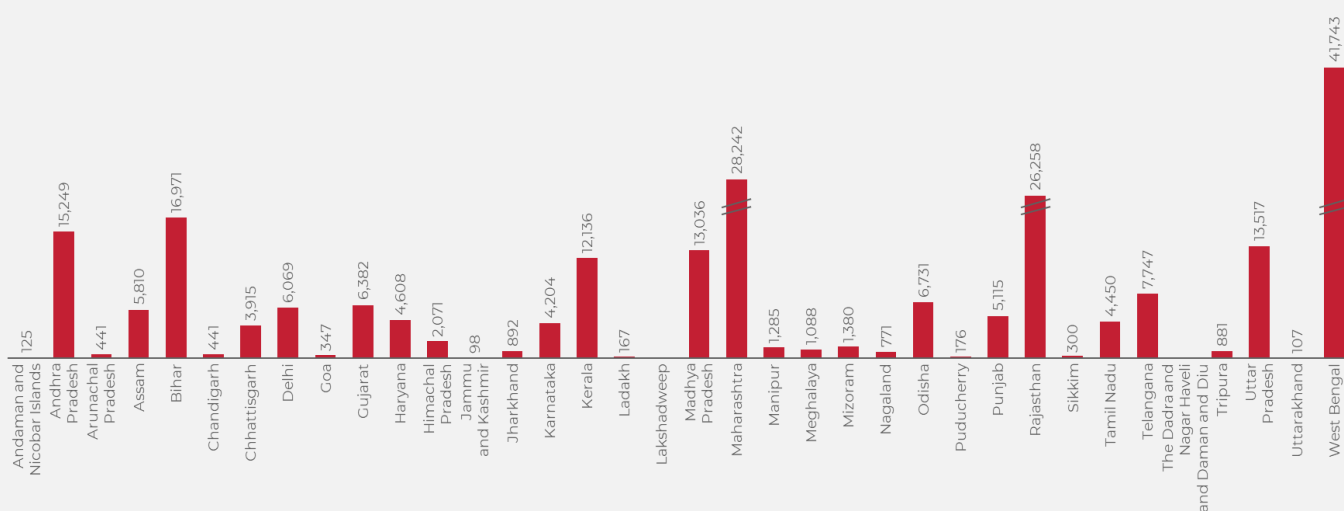
⁷ University Grants Commission, Guidelines for Internationalisation of Higher Education in India, 2021

Most teacher recruitments in various central and state universities adhere to the university's guidelines for rules, regulations, salaries, and employment terms, and they do not fall under civil service provisions.⁸ In this scenario, the university announces job openings, forms a selection committee, and requests candidates to participate in interviews. The standard selection procedure across most central and state universities in the nation tends to be time-consuming. In many state universities, appointments are directed towards civil service positions, where teachers are employees of the state government.

Teacher training institutes have reduced over the past decade:

The total number of Teacher Training institutions as per AISHE have reduced from 4895 in 2012-13 to 3,301 in 2021-22.⁹ The total number of students enrolled in these institutes has also dropped, from 3.29 lakh in 2012-13 to 2.33 lakh in 2021-22.¹⁰ The share of women enrolled in these institutes has largely remained the same since 2012-13 at around 63%.¹¹

Figure 3: Enrolment in teacher training by state and UT (2021-22)



Source: Ministry of Education, All India Survey of Higher Education, 2021-22

Research, innovation, and entrepreneurship

This includes fostering a robust research ecosystem in HEIs by inclusion of research and internships in the undergraduate curriculum, promotion of faculty career management systems that give due weightage to research, governance and regulatory changes that encourage an environment of research and innovation, and the like.

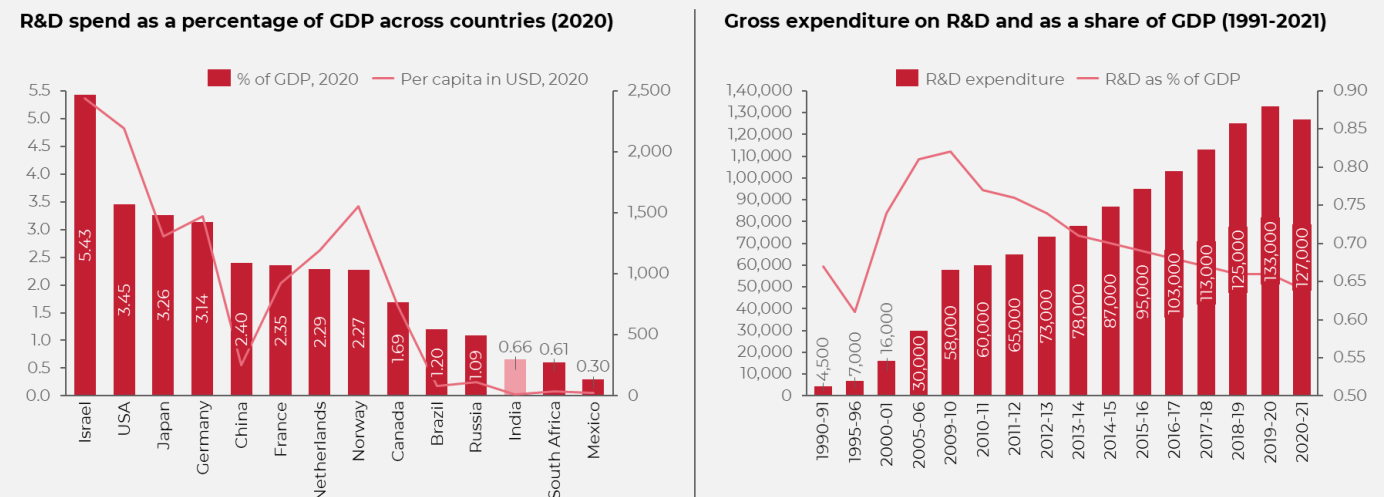
⁸ Centre for Policy Research in Higher Education (CPRHE), Teacher Recruitment in Higher Education in India - An Analysis of National Eligibility Test (NET) Results, 2017

⁹ Ministry of Education, All India Survey of Higher Education, 2021-22, Ministry of Education, All India Survey of Higher Education, 2012-13

¹⁰ Ministry of Education, All India Survey of Higher Education, 2021-22, Ministry of Education, All India Survey of Higher Education, 2012-13

¹¹ Ministry of Education, All India Survey of Higher Education, 2021-22

Figure 4: R&D spending as a percentage share of GDP across countries

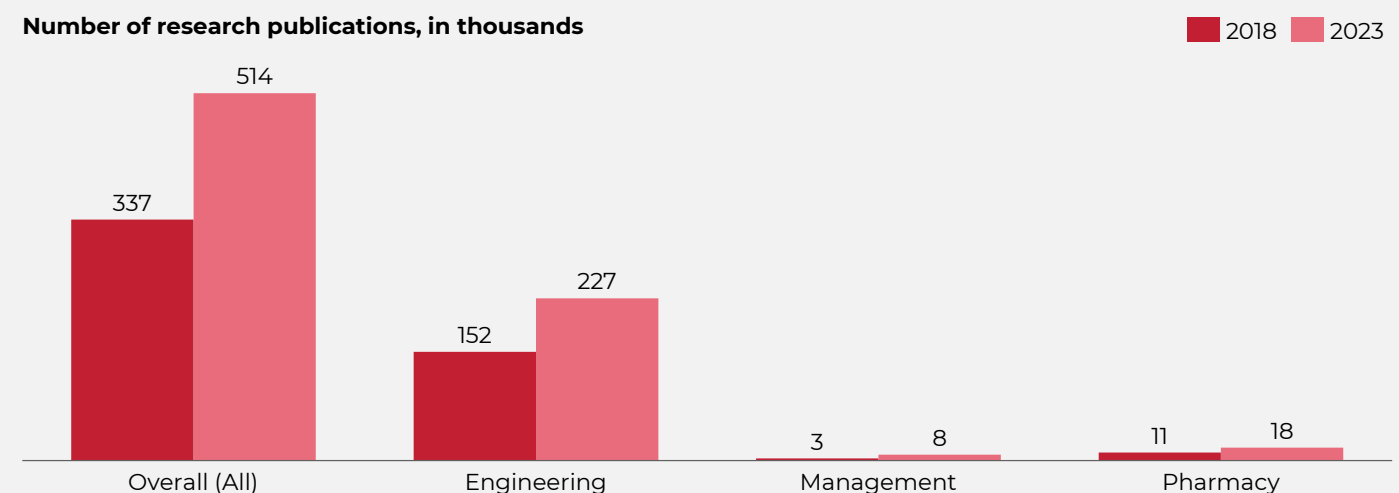


Source: Department of Science and Technology, R&D Statistics at a Glance, 2022-23

R&D investments, personnel, and associated research output have seen a rise:

India's gross expenditure on R&D (GERD) has consistently been increasing over the years, tripling between 2008 and 2018. Similarly, the number of researchers per million has doubled since 2000, with India occupying the third rank in the number of Ph.D.s awarded in science and engineering after the USA and China. Additionally, India's share in global research publication output has increased over the years, with the country being ranked third in scientific publications as per the National Science Foundation (NSF) database.¹²

Figure 5: Number of research publications of India



Source: Ministry of Education, India Rankings 2018 and 2023

While there has been progress, quality of research in India needs to improve further:

As of 2022, India ranked third globally in terms of publications and citations with more than 200,000 publications, as compared to fifth rank in 2016.¹³ However, the production of high-quality research at the university and college-level remains a challenge. A 2019 UGC report states that, in many instances, theses or dissertations written by higher education students do not conform with international standards and

¹² Ministry of Science & Technology, India's R&D expenditure & scientific publications on the rise, 2020

¹³ PIB, PM Shri Narendra Modi to inaugurate Akhil Bhartiya Shiksha Samagam on 3rd anniversary of National Education Policy on 29th July in New Delhi, 2023

do not add significant value to the applied aspects of respective disciplines.¹⁴ This issue has been identified, and the government has taken several measures, such as establishing the NRF. The effect of such measures is evident in the improving global ranking of Indian HEIs. For instance, since 2015, the number of Indian universities in the QS Rankings has increased by 318% in 2024, reaching a total of 45.¹⁵

Anusandhan National Research Foundation aims to cultivate a culture of R&D and innovation:

In line with NEP's guidance to set up an apex body for strengthening the research ecosystem in the country, the Anusandhan National Research Foundation Bill, 2023 was introduced in Lok Sabha on August 4, 2023.¹⁶ The Bill provides for establishing the Anusandhan National Research Foundation (NRF) supported by a financial allocation of Rs. 50,000 crores, which will initiate, nurture, and support research activities at academic institutions with specific focusing on universities and colleges, where such research capabilities are still nascent. This body will improve linkages across stakeholders in India's research ecosystem and will provide a platform for greater participation of industries and government. Furthermore, NRF will subsume the Science and Engineering Research Board (SERB) as part of its expanded ambit of functions.¹⁷

Recent policy efforts focus on improving R&D capabilities and providing financial support to current and aspiring researchers:

The recently launched State University Research Excellence scheme by the Science and Engineering Research Board (now, subsumed under Anusandhan National Research Foundation) supports R&D in cutting-edge fields like science and by assisting active researchers from state and private universities and colleges throughout India.¹⁸ A similar effort by the UGC is the push to establish Research and Development Cells in HEIs in order to create a robust mechanism for development of, and strengthening R&D within HEIs. Another notable scheme is the Innovation in Science Pursuit for Inspired Research (INSPIRE) program, which aims to enhance India's skilled human resources in the R&D field. The program awards scholarships to students for pursuing sciences at the undergraduate level and offers fellowships to students and faculty for pursuing Ph.D. and post-doctoral research, respectively.¹⁹

India's incubation ecosystem is supported by various central ministries:

The number of incubators and accelerators in India have grown 7 times between 2015 and 2022, which has supported India in becoming the third largest startup ecosystem in the world.²⁰ Operating under the NITI Aayog, the Atal Innovation Mission is a key initiative that sets up state-of-the-art incubators called Atal Incubation Centres (AICs) at various universities, institutions, and corporations.²¹ Other noteworthy incubation initiatives include the Technology Business Incubators (TBI) established by the Department of Science and Technology (DST) which fosters startups that work on new technology, knowledge, and innovation.²² The Department of Biotechnology has introduced BioNest to amplify biotechnology-based incubation in the country.²³ Additionally, the Ministry of Electronics and Information Technology (MeitY) has initiated the TIDE program, that offers both financial technical support to incubators supporting ICT startups.²⁴ Furthermore, various state-level initiatives such as iStart Nest by Government of Rajasthan, Kerala Startup Mission (KSUM) by Government of Kerala, and Andhra Pradesh Innovation Society (APIS) by the Government of Andhra Pradesh, promote the growth of entrepreneurship and create an ecosystem to support startup businesses.

Equity and Inclusion

14 University Grants Commission, Promoting and Improving the Quality of Research in Indian Universities/Colleges, 2019

15 Prime Minister's Office, Prime Minister reiterates his commitment for qualitative changes in the education sector, 2024

16 PRS Legislative Research, Anusandhan National Research Foundation Bill, 2023

17 Ministry of Education, National Research Foundation to strengthen the research ecosystem in the country, 2021; Ministry of Science & Technology, Cabinet approves Introduction of National Research Foundation Bill, 2023 in Parliament to strengthen research eco-system in the country, 2023

18 Ministry of Science & Technology, New scheme launched to augment research capabilities in state universities and colleges, 2022

19 Ministry of Science & Technology, Interaction of Dr. Jitendra Singh Hon'ble Minister of State (Independent Charge) of the Ministry of Science and Technology & Ministry of Earth Sciences with INSPIRE Beneficiaries, 2022; Ministry of Science & Technology, Press release on the INSPIRE programme, 2021; University Grants Commission, Guidelines for Establishment of Research & Development Cell In Higher Education Institutions, 2022

20 Invest India, The Indian Unicorn Landscape, 2022; Ministry of Commerce and Industry, Startup India, accessed December 2023

21 Government of India, Atal Incubation Centre, Accessed July 2023; Ministry of Micro, Small & Medium Enterprises, MSME Innovative Scheme under MSME Champions Scheme being implemented for developing and nurturing innovative ideas, 2023

22 Startup India, Incubators, Accessed August 2023

23 Startup India, Incubators, Accessed August 2023

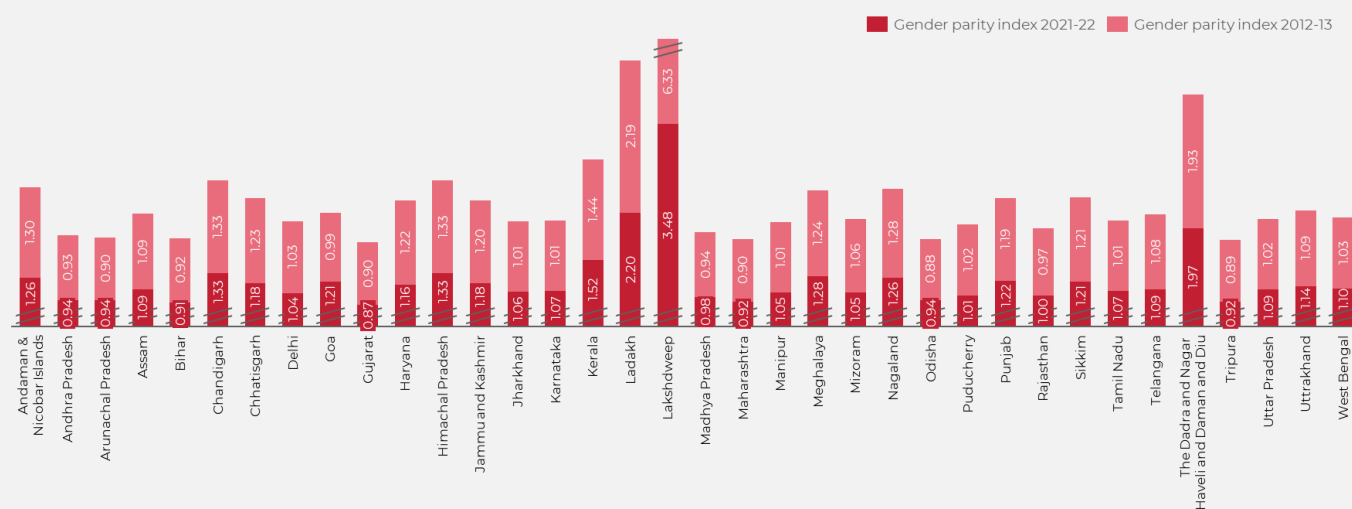
24 Startup India, Incubators, Accessed August 2023

Entry into quality higher education can open a vast array of possibilities that can lift both individuals as well as communities out of the cycles of disadvantage. India has come a long way in enabling improved higher education access for women, disadvantaged social groups and minority communities.

Gender parity has improved in higher education enrolments:

India has been able to achieve gender parity in higher education enrolments, with about 48% female students enrolled in HEIs in 2022, up from 45% in 2012.²⁵ In 2021-22, the GER for females was 28.5, more than the GER of males, which was at 28.3.²⁶ In M.Phil. and Post Graduate programs, the enrolment of female students surpasses that of male students, with about 75% female students in M.Phil. programs and 55% in Post Graduate programs.²⁷

Figure 6: Gender parity index in higher education (2012-13 and 2020-22)

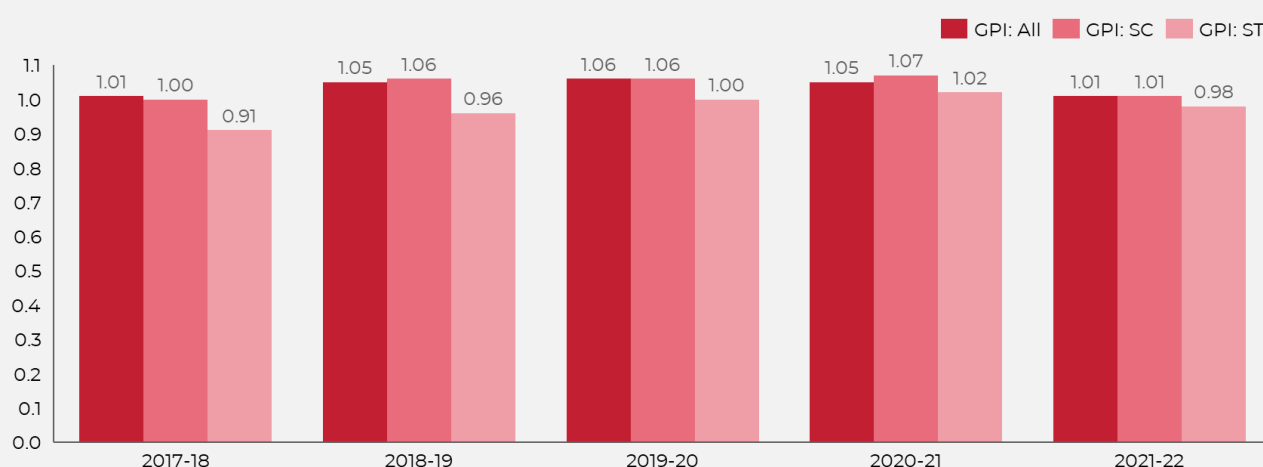


Source: Ministry of Education, All India Survey of Higher Education 2012-13 and 2021-22

Enrolment of socio-economically disadvantaged groups (SEDGs) in HEIs has increased since 2014:

Between 2014-15 to 2021-22, the increase in student enrolment in SC, ST and OBC categories has been approximately 44%, 65%, and 45% respectively.²⁸

Figure 7: Gender parity index and gross enrolment ratio (2017-22)

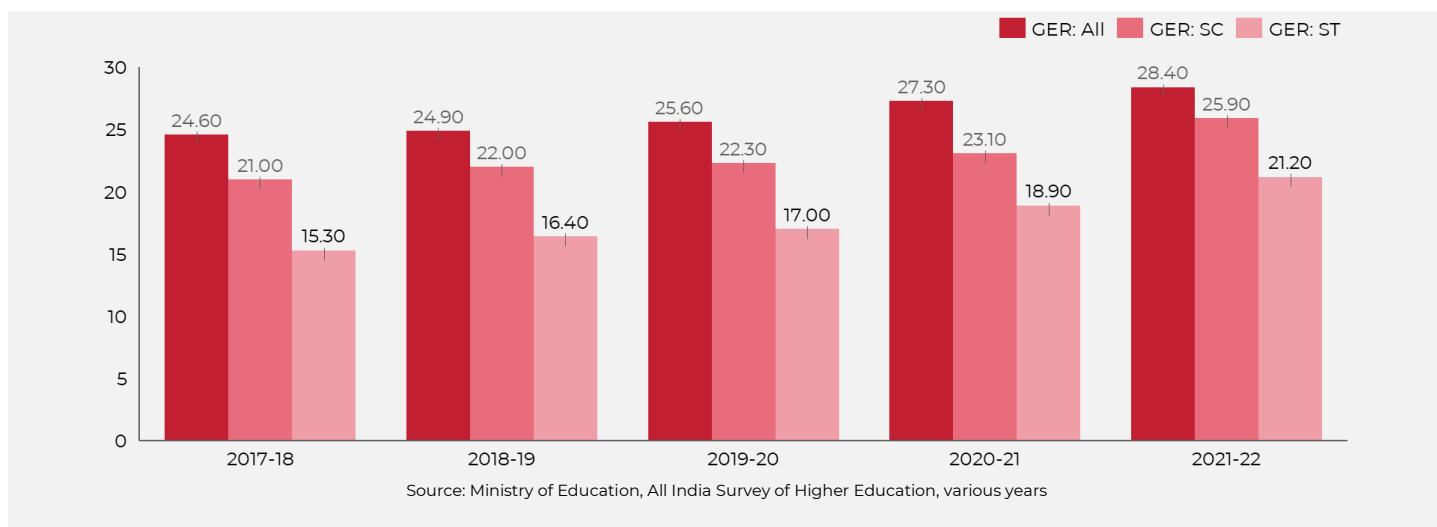


²⁵ Ministry of Education, All India Survey of Higher Education, 2021-22; Ministry of Education, All India Survey of Higher Education, 2012-13

²⁶ Ministry of Education, All India Survey of Higher Education, 2021-22

²⁷ Ministry of Education, All India Survey of Higher Education, 2021-22

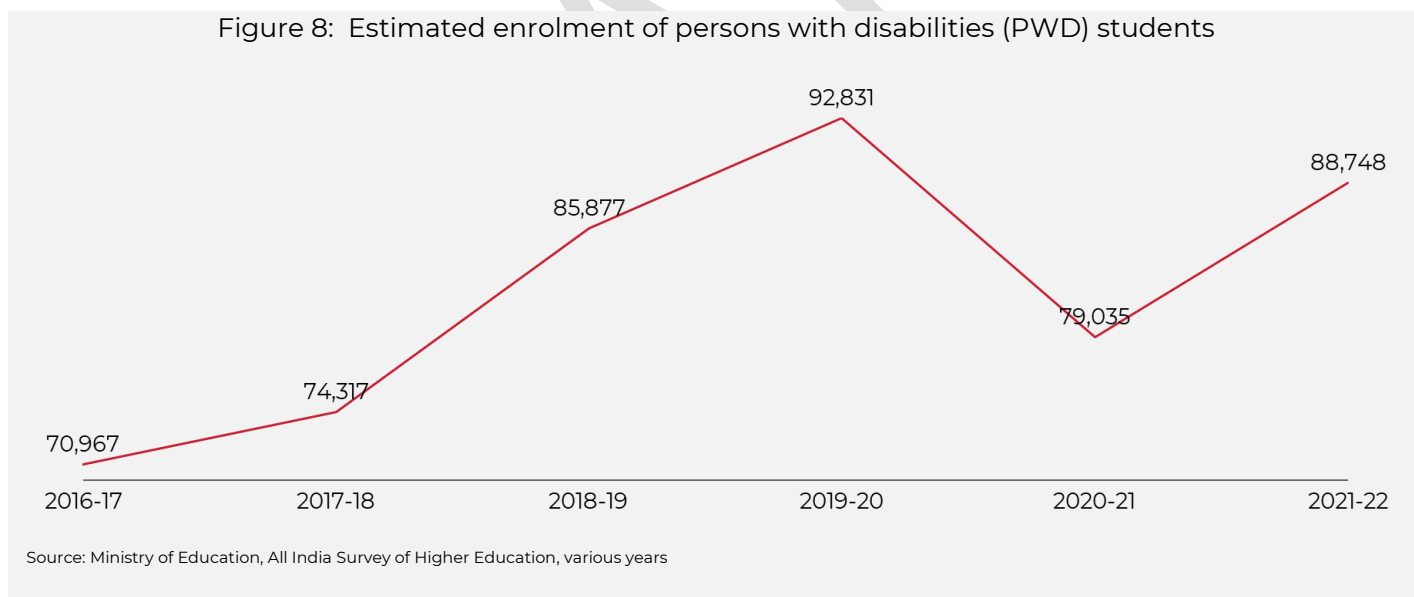
²⁸ Ministry of Education, All India Survey of Higher Education, 2021-22



Increasing scholarships and improving educational infrastructure for PwDs are key policy priorities:

During the period from 2013 to 2021, there has been an increase in budget allocations for scholarship programs tailored for Persons with Disabilities (PwDs), going from INR 8.6 lakhs in 2013-14, to INR 555 crores in 2021-22.²⁹ These programs encompass pre-matric, post-matric, overseas education, national fellowships, and free coaching. The government also plans to further extend scholarships initiated by private and philanthropic universities, targeting students from disadvantaged backgrounds.³⁰

Figure 8: Estimated enrolment of persons with disabilities (PWD) students



Vocational education within higher education

The NEP proposes the integration of vocational education into all HEIs in a phased manner over the next decade. Focus areas for vocational education will be chosen based on skills gap analysis and mapping of local opportunities.

Recent policy efforts aim to promote skill-based vocation education in HEIs:

To enhance the availability of vocational and skill-based education, the UGC has authorised HEIs to offer credit-based programs across levels such as Certificate, Diploma, B. Voc, and M. Voc., including multiple entry-exit options. Additionally, the UGC has provided guidelines for HEIs to introduce apprenticeship and internship-embedded degree programs. Finally, the National Apprenticeship Training Scheme offers

²⁹ Department of Empowerment of Persons with Disabilities, Scheme wise and year wise amount released and number of beneficiaries, 2013-14 to 2020-21
³⁰ Ministry of Education, National Education Policy, 2020

practical exposure in professional settings to technically qualified individuals, including engineering graduates, diploma holders, and vocational course graduates.³¹ The All India Council for Technical Education has also released the National Vocational Education Qualification Framework (NVEQF) to incorporate vocational education within existing traditional educational pathways in both school and higher education domains.³²

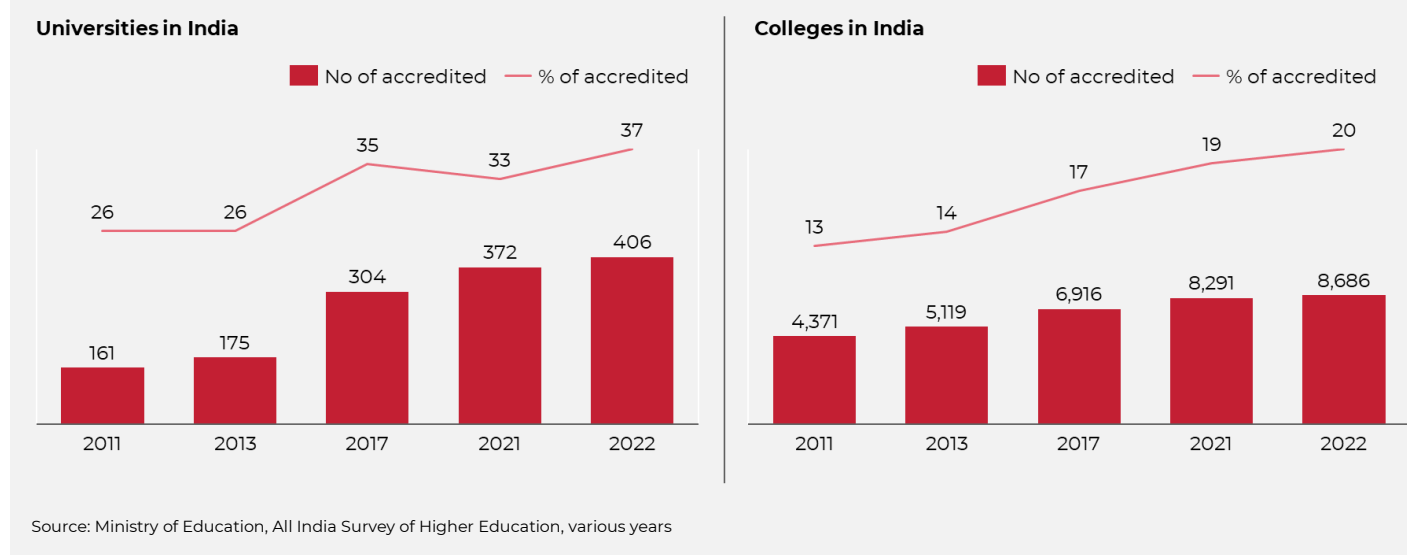
National educational targets include integrating vocational education into mainstream learning and increasing industry exposure of students:

By 2025, the government aims to ensure that at least 50% of learners within the school and higher education system are exposed to vocational education. The government is also planning to mandate that HEIs offer vocational education, either independently or in collaboration with industry and non-governmental organisations (NGOs).³³

Resourcing, Governance, and Assessment of higher education

India has made notable progress in digitising classrooms and is actively working towards cultivating industry-ready graduates while promoting greater inclusivity and flexibility for learners from diverse backgrounds.

Figure 9: Number and share of accredited universities and colleges in India



Promotion of digitalisation has received significant policy attention in the last decade:

The availability of computer centres in universities has increased from 88% in 2012-13 to 93% in 2021-22. The connectivity of universities to critical new-age digital education platforms such as National Knowledge Network and National Mission on Education through Information and Communication Technology (NMEICT) sits at 56% and 41% respectively.³⁴ Study Webs of Active-Learning for Young Aspiring Minds (SWAYAM) is a national massive open online course platform for lifelong and online learning. It allows students the opportunity to use certifications earned from the platform for credit transfer to the universities where they are enrolled.³⁵ As of December 2023, more than 9 lakh students have received certificates from proctored exams through SWAYAM.

Separately, ICT initiatives like Smarter Automation Engine for Universities (SAMARTH) seek to support HEIs with their administrative processes and manage students throughout their university careers.

31 PRS Legislative Research, Demand for Grants 2023-24 Analysis – Education, 2023

32 All India Council for Technical Education, National Vocational Education Qualification Framework (NVEQF), Accessed August 2023

33 National Council of Educational Research & Training, Background Note on Reimagining Vocational Education and Skill-building, 2021; UNESCO, State of the Education Report for India - Technical and Vocational Education and Training (TVET), 2020

34 Ministry of Education, All India Survey of Higher Education, 2021-22; Ministry of Education, All India Survey of Higher Education, 2012-13

35 Ministry of Education, Digital India initiative of Government has revolutionized education access in rural areas, 2023

Approximately 2,700 HEIs across 32 States and UTs were using this e-governance platform as of December 2023.³⁶

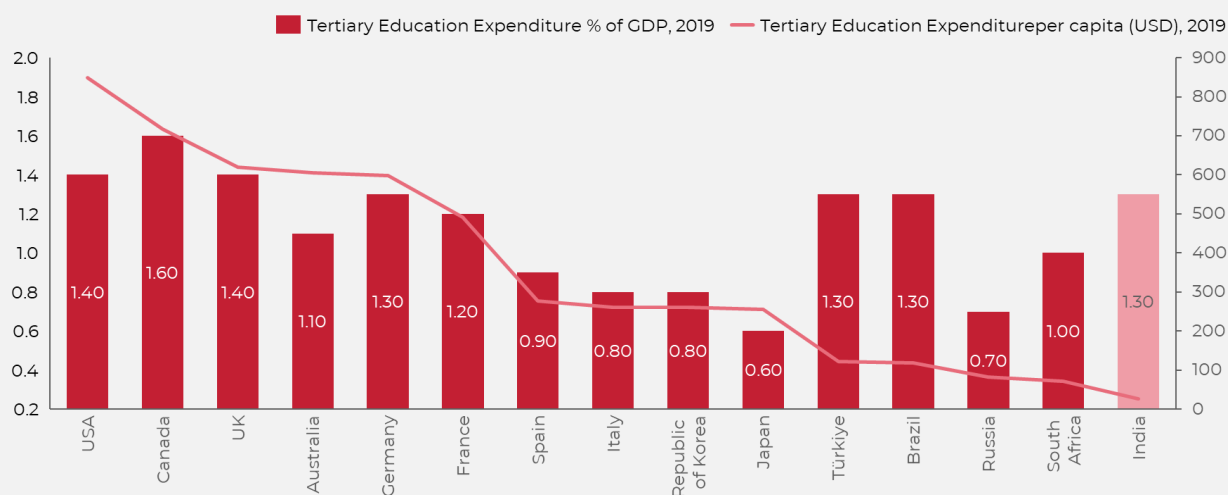
Recent government efforts have focused on improving flexibility in learning and instilling industry-ready skills in graduates:

The Academic Bank of Credits (ABC), set up by the Ministry of Education and the Ministry of Electronics and IT in 2021 enables students to transfer their credits from one program to another, allowing them the flexibility to study at various institutions while pursuing a degree or diploma.³⁷ Till July 2023, about 1 crore ABC student IDs have been created and about 1400 universities and institutions are registered on the portal.³⁸ Additionally, the UGC released a National Higher Education Qualifications Framework in 2023 aiming to improve employability of graduates. It offers a competency-based skills framework, connecting the learning outcomes of various educational programs (including vocational, technical, and general) to allow learners to advance to different levels regardless of their initial educational background. The framework aligns itself with skills required for specific job roles in various sectors.³⁹ In line with this, the UGC also issued guidelines in 2023 for establishing a "Sustainable and Vibrant University-Industry Linkage System." These guidelines emphasize strengthening university-industry collaboration to align curricula with industry needs, enhance research and innovation, and provide practical training opportunities to students.⁴⁰

Financing for Higher Education

Affordable higher education is crucial for fostering equal opportunity, social mobility, economic growth, and global competitiveness. Over the years, the budget for higher education and the number of scholarships for students have increased.

Figure 10: Tertiary education expenditure as a share of GDP and per capita (USD) for India and various countries (2019)



Source: UNESCO

Public expenditure on higher education has increased:

The Union Budget allocation to the Department of Higher Education for FY 2023-24 reached an all-time high of INR 44,094 crore. This reflects an 8% increase in the budget allocation for the Department of Higher Education compared to the FY 2022-23.⁴¹ However, India significantly lags behind in the per-capita

³⁶ PIB, PM Shri Narendra Modi to inaugurate Akhil Bhartiya Shiksha Samagam on 3rd anniversary of National Education Policy on 29th July in New Delhi, 2023

³⁷ Ministry of Education, About - Academic Bank of Credits (ABC), Accessed July 2023

³⁸ Ministry of Education, Academic Bank of Credits (ABC), Accessed July 2023

³⁹ University Grants Commission, Draft National Higher Education Qualifications Framework (NHEQF), 2022

⁴⁰ University Grants Commission, Guidelines on Sustainable and Vibrant University-Industry Linkage System for Indian Universities, 2023

⁴¹ Ministry of Education, Ministry of Education has been allocated Rs 112899.47 crore, the highest allocation ever to the Education Ministry – Shri Dharmendra Pradhan, 2023

spent on tertiary education (26 USD) as compared to developed countries such as USA (840 USD) and UK (860 USD).⁴²

Over two-thirds of public spending on university and higher education comes from states and UTs:

More than two-thirds of all universities in India are under the state governments. As per AISHE 2021-22, out of the 685 government-managed universities, 35% (240) are central universities, and 64% (445) are state universities.⁴³ As per 2020-21 budgetary estimates, expenditure by states and UTs stood at 68% of the total public spending (i.e., expenditure by states and UTs as well as the centre) on university and higher education.⁴⁴

Number of scholarships and loans for higher education have grown:

The number of scholarships granted to students at universities and colleges under the Pradhan Mantri Uchchatar Shiksha Protsahan Yojna - Central Sector Scheme for Scholarship have grown from about 1.2 lakh in 2017-18 to 1.8 lakh in 2022-23.⁴⁵ As per the RBI, education loans have grown substantially over the past decade, from INR 49,800 crores of gross bank credit outstanding in FY11 to INR 84,677 crores in FY21.⁴⁶

⁴² UNESCO, UNESCO Data, Accessed 2024

⁴³ Ministry of Education, All India Survey of Higher Education, 2021-22

⁴⁴ Ministry of Education, Analysis of Budgeted Expenditure on Education: 2018-19 to 2020-21, 2022

⁴⁵ Lok Sabha Unstarred Question No. 2775, Financial Assistance to Meritorious Students under PM-USP CSSS, 2023

⁴⁶ RBI, Report on Trend and Progress of Banking in India 2012-13, 2013; RBI, Report on Trend and Progress of Banking in India 2021-22, 2022

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by adopting effective higher education practices. These revolve primarily around holistic and equitable learning, robust professional development of teachers, strong research capacity, and improved governance.



Greater multidisciplinary learning:

Currently, there exists a strict division among academic fields in HEIs, involving early specialisations and the categorisation of students into specific and limited subjects of study.⁴⁷ Establishing large HEIs, via clustering, that promote interdisciplinary learning and innovative pedagogies such as internships, community service, and skill-based courses can ensure holistic education that combines theoretical knowledge with real-world applications.⁴⁸ There is also scope to strengthen the vocational education curriculum by facilitating collaborations between HEIs, industry, and NGOs.⁴⁹



Higher professional development avenues for faculty:

Referring to teaching as the “most crucial profession”, the World Bank says that effective teacher development requires making the profession attractive through communication strategies and financial incentives, using apt hiring, recognition, and reward policies, and providing requisite skill-building opportunities, alongside continuous support.⁵⁰ In India, systems for merit-oriented career advancement and progression are limited.⁵¹ By offering attractive incentives such as greater administrative and academic autonomy and career progression based on teaching excellence, impactful research, and dedicated service, HEIs can attract and retain top-tier educators.⁵²



Robust research capabilities of HEIs:

According to AISHE 2021-22, only just ~0.5% of enrolled students in HEIs were pursuing Ph.D. courses, which are critical to HEI's research ecosystem.⁵³ Providing seed funding for smaller HEIs' research initiatives and establishing high-quality peer-review processes across disciplines can nurture a culture of research and innovation in India's HEIs.⁵⁴



Increase in access, equity, and inclusion in HEIs:

Out of the total enrolment in 2021-22, approximately 15% belonged to the Scheduled Castes group, 6% were from the Scheduled Tribes group, and 38% were from the OBC category.⁵⁵ Expanding opportunities for excellent public education in rural areas, scholarships for disadvantaged students, delivering curriculum in local languages, embracing online education and Open Distance Learning (ODL) options, and ensuring that all learning materials and infrastructure are easily accessible and accommodative for learners with disabilities can greatly improve access to education.⁵⁶



Improvement in governance of HEIs:

Inadequate governance and an ineffective regulatory system are key administrative challenges in the context of HEIs.⁵⁷ The higher education ecosystem can be better governed by onboarding qualified independent boards to conduct accreditation and assessments of HEIs across the country, establishing a single regulator, and clustering small HEIs into large multidisciplinary universities.⁵⁸

47 Ministry of Education, National Education Policy, 2020

48 Ministry of Education, National Education Policy, 2020

49 Ministry of Education, National Education Policy, 2020

50 World Bank, Successful Teachers, Successful Students: Recruiting And Supporting Society's Most Crucial Profession, 2021

51 Ministry of Education, National Education Policy, 2020

52 Ministry of Education, National Education Policy, 2020

53 Ministry of Education, All India Survey of Higher Education, 2021-22

54 Ministry of Education, National Education Policy, 2020

55 Ministry of Education, All India Survey of Higher Education, 2021-22

56 Ministry of Education, National Education Policy, 2020

57 Ministry of Education, National Education Policy, 2020

58 Ministry of Education, National Education Policy, 2020



Greater use of digital education and e-learning platforms:

The availability of computer centres in universities has increased from 88% in 2012-13 to 93% in 2021-22.⁵⁹ However, there is an opportunity to increase the connectivity of universities to critical new-age digital education platforms such as the National Knowledge Network and National Mission on Education through Information and Communication Technology.⁶⁰



Greater internationalisation of higher education:

India's current higher education curricula and teaching pedagogies do not match global standards.⁶¹ HEIs can collaborate with faculty in leading international universities to add a global perspective in Indian curricula, enter into twinning agreements or semester abroad programs that allow Indian students to complete a part of their degree in another country at a partnering institution, and promote faculty exchange programs to increase faculty exposure to foreign curriculum and pedagogies.⁶²

Challenges

This section highlights some of the major challenges and constraints facing the higher education ecosystem and HEIs in India. Efforts to resolve such challenges will ensure better higher employability, more efficient processes and governance, greater research orientation, greater course diversity and adequate financing.



Inadequate employable skills training within the existing HEI curriculum:

The India Skills Report 2018 by the UNDP found the overall employability across higher education disciplines to be about 45%.⁶³ Several industry surveys conducted in recent years have revealed varying levels of employability among disciplines like engineering graduates, ranging from 10% to 40%, depending on the specific job role.⁶⁴ As per the surveys, existing curriculum places limited emphasis on critical soft skills, particularly communication skills, which are highly valued by employers.



Inefficient faculty recruitment process:

According to the Standing Committee on Human Resource Development, the recruitment process for faculty in HEIs is lengthy and burdened with excessive procedural formalities, creating a shortage of qualified faculty in HEIs.⁶⁵ Additionally, the UGC-NET examination and the process of sanctioning faculty positions are time-consuming.⁶⁶ Limited incentives to join and remain in the higher education teaching profession, such as lack of research and post-graduate scholarships and higher salaries, is an additional reason for high vacancies.⁶⁷



Limited availability of qualified PhD scholars and research faculty:

Despite the number of PhD enrolments doubling over the last five years, its proportion to total enrolment has fallen.⁶⁸ Additionally, the completion rate of PhDs in India is low, as evidenced by how only about 12% of students enrolled at the PhD level were awarded degrees.⁶⁹ These trends result in shortages in trained faculty. This also contributes to a small number of emerging researchers, hampering India's overall R&D capabilities.⁷⁰

59 Ministry of Education, All India Survey of Higher Education, 2021-22; Ministry of Education, All India Survey of Higher Education, 2012-13

60 Ministry of Education, All India Survey of Higher Education, 2021-22; Ministry of Education, All India Survey of Higher Education, 2012-13

61 Ministry of Education, National Education Policy, 2020; University Grants Commission, Guidelines for Transforming HEIs into Multidisciplinary Institutions, 2022

62 University Grants Commission, Guidelines for Internationalisation of Higher Education in India, 2021

63 UNDP, India Skills Report, 2018

64 Brookings, Reviving Higher Education in India, 2019

65 Standing Committee on Human Resource Development, Issues and challenges before higher educational sector in India, 2017

66 Standing Committee on Human Resource Development, Issues and challenges before higher educational sector in India, 2017

67 Brookings, Reviving Higher Education in India, 2019

68 Ministry of Education, AISHE Reports, multiple years

69 Ministry of Education, All India Survey of Higher Education, 2021-22

70 Brookings, Reviving Higher Education in India, 2019



Inferior perception of vocational education programs:

Vocational education is commonly seen as inferior compared to conventional mainstream education, often associated with students who may struggle with the latter.⁷¹ This perception influences the decisions that students ultimately make regarding enrolling in vocational training programs. Furthermore, teachers and school management often lack the necessary know-how and skill to integrate and provide vocational education.⁷²



Excessive fragmentation of HEIs leading to complexity in governance:

In India, there are approximately 300 affiliating universities, and these universities collectively have about 40,000 colleges affiliated to them.⁷³ Due to this, university administrators face high administrative pressure and are unable to enforce the quality standards necessary for establishing and running colleges.⁷⁴



Insufficient financing options for quality higher education:

About 69% of universities in India are privately owned (unaided), and they account for around 26% of total student enrolments.⁷⁵ According to the 75th Round of the National Sample Survey, private higher education is expensive, with the average annual expenditure in unaided private institutions to pursue a basic course at the graduate level at approximately INR 20,000, compared to about INR 10,000 in government-run institutions.⁷⁶

⁷¹ Ministry of Education, National Education Policy, 2020

⁷² National Council of Educational Research & Training, Background Paper on Reimagining Vocational Education, 2020

⁷³ Ministry of Education, All India Survey of Higher Education, 2018-19

⁷⁴ Brookings, Reviving Higher Education in India, 2019

⁷⁵ Ministry of Education, All India Survey of Higher Education, 2021-22

⁷⁶ Ministry of Statistics and Programme Implementation, Household Social Consumption on Education in India - NSS 75th Round, 2017-18

Institutional framework and policy landscape –

Key ministries, departments and institutions

This section highlights the major central government bodies and national-level institutions that oversee higher education.



Ministry of Education⁷⁷

The Ministry of Education ensures the implementation of the National Education Policy. It houses the Department of Higher Education, which is responsible for the overall development of the basic infrastructure of the Higher Education sector, encompassing both policy and planning. The Department oversees the expansion of access and the qualitative improvement of Higher Education, fostering the establishment of world-class universities, colleges, and other institutions.



University Grants Commission (UGC)⁷⁸

The UGC provides grants to eligible universities and colleges, and advises the central and state governments on measures necessary for the development of higher education in India.



All India Council for Technical Education (AICTE)⁷⁹

The AICTE grants approval for new technical institutions, courses, and intake variations, ensuring quality through accreditation and setting norms and standards. It promotes technical education through schemes for women, the handicapped, and weaker sections, and fosters innovation, faculty development, and research.



Higher Education Financing Agency (HEFA)⁸⁰

HEFA, a joint venture of the Ministry of Education, Government of India, and Canara Bank, finances the creation of capital assets in premier educational institutions to develop top-ranked institutions like IITs, NITs, IISCs, and AIIMS. Its scope includes school education and institutions under the Ministry of Health, focusing on improving academic and infrastructure quality to achieve global rankings.



National Assessment and Accreditation Council (NAAC)⁸¹

NAAC conducts assessment and accreditation of Higher Educational Institutions (HEI) such as colleges, universities or other recognised institutions to derive an understanding of the 'Quality Status' of the institution.

Major policies and programs

This section highlights existing policies related to higher education and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI for States Platform.



National Education Policy, 2020⁸²

The National Education Policy aims to increase the Gross Enrolment Ratio in higher education, including vocational education, from 26.3% (2018) to 50% by 2035.

The policy proposes various reforms in higher education through:

⁷⁷ Ministry of Education, About Department Of Higher Education, Accessed 2023

⁷⁸ University Grants Commission, About UGC, Accessed 2023

⁷⁹ Department of Higher Education, All India Council for Technical Education (AICTE), as accessed on June 2024

⁸⁰ HEFA, About Us, Accessed 2024

⁸¹ NAAC, Welcome to NAAC, Accessed 2023

⁸² Ministry of Education, National Education Policy, 2020

- Envisioning the transformation of higher education institutions into expansive, multidisciplinary universities, colleges, and HEI clusters/Knowledge Hubs, each with a target of accommodating 3,000 or more students
- Setting up a National Research Foundation (NRF) that will provide a reliable base of merit-based research funding, foster a culture of research by introducing suitable incentives for recognition of outstanding research, and undertake major initiatives to catalyse and grow research at State Universities and other public institutions where research capabilities are currently constrained
- Establishing the Higher Education Commission of India (HECI), serving as an overarching institution to guarantee that distinct functions such as regulation, accreditation, funding, and academic standard setting are carried out by separate, autonomous, and empowered bodies
- Strengthening Open and Distance Learning (ODL) Programmes to facilitate inclusive learning for all students, with special emphasis on Socio-Economically Disadvantaged Groups (SEDGs)
- Setting up of an International Students Office at each HEI hosting international students to facilitate research/teaching collaborations and faculty/student exchanges with esteemed foreign institutions



National Apprenticeship Training Scheme (NATS)

Through the scheme, the government aims to enhance the employability of students and provides apprenticeship training by industry and commercial organisations. Under the scheme which will run till 2026, apprentices completing graduate and diploma programme in Engineering, Humanities, Science and Commerce will be awarded a stipend of Rs. 9,000 and Rs. 8,000/- per month respectively.⁸³



National Credit Framework⁸⁴

In line with the NEP 2020's vision to make education more holistic and integrated, the National Credit Framework integrates credits earned by students across school, higher, vocational, and skill education by unifying and enabling mobility across the National Higher Education Qualification Framework, the National School Education Qualification Framework, and the National Skills Qualification Framework. Students earning academic or other credits through any of these frameworks can transfer and accumulate the same across institutions by digitally storing and accessing them through the Academic Bank of Credit.



Pradhan Mantri Uchchatar Shiksha Abhiyan (PM-USHA)⁸⁵

PM-USHA is a centrally sponsored scheme launched to improve access, equity, and excellence in state higher education systems. It builds on the previous RUSA initiatives and focuses on enhancing graduate employability, promoting market-linked courses, industry connections, skill-based education, and leveraging technology for better access and quality. The scheme supports the accreditation of institutions, adoption of the National Curriculum Framework, and improvement of digital infrastructure. Aligned with the National Education Policy 2020, PM-USHA prioritizes inclusive education, gender inclusion, and catering to socio-economically disadvantaged groups to achieve higher Gross Enrolment Ratios and better educational outcomes.

⁸³ Ministry of Education, Cabinet approves continuation of National Apprenticeship Training Scheme for next five years, 2021

⁸⁴ UGC, National Credit Framework, 2023

⁸⁵ Department of Higher Education, PM-USHA

Way forward

India's goals of emerging as a knowledge superpower, driving inclusive innovation, and advancing rapid economic and social growth depend significantly on the strength of India's higher education system. This system must meet the aspirations of the 10 million-plus young graduates entering the workforce annually, do so affordably and equitably, generate globally-competitive research, and create world-class, efficiently-run institutions.

Higher education policies and programs following the NEP carry several components critical for progress: an emphasis on student centricity and multidisciplinary learning, institutional consolidation, motivated faculty, well-integrated vocational education, international mobility, digital learning, and industry partnerships. Challenges of regulation, building deeper industry linkages, and capacity gaps across teachers and administrators require continued attention.

Some ideas to enable and sustain policy progress across these fronts, based on proposals from institutions like the NITI Aayog, industry bodies like FICCI, and development partners like UNESCO and the World Bank, include:

 Continue enhancing quality assurance	Encourage clustering of HEIs to create multidisciplinary institutes⁸⁶ - Strengthen institutions by adding departments in varied subject areas like Languages, Philosophy, Art, Pure and Applied Sciences, and other subjects as needed for a multidisciplinary institution - In line with UGC Guidelines, merge single-stream institutions with other multidisciplinary institutions under the same or different management(s)	Strengthen accreditation systems and capacity⁸⁷ - Encourage the growth of third-party, credible accreditation agencies empanelled through stringent processes - Build the capacity of independent accreditation and qualification agencies by supporting adequate staffing and adopting an outcome-based approach that effectively marshals system-level performance data for rapid assessments
 Enable greater research and innovation	Incentivise the quality of research output⁸⁸ Enhance policy guidelines that ensure advances on key research quality and impact metrics such as citations in top international journals and granted patent applications through rigorous peer review and ethics policies as well as international research collaborations.	Enhance research financing⁸⁹ Enable industry, policy, and civil society linkages (e.g., through corporate endowments, industry collaboration wings within universities and collaborations with on-campus incubators and accelerators) to advance market, public, and philanthropic support, (including with the Higher Education Financing Agency), and involvement in the creation of high-quality and relevant academic research across disciplines.

⁸⁶ UGC, Guidelines for Transforming Higher Education Institutions into Multidisciplinary Institutions, 2022

⁸⁷ NITI Aayog, Strategy for New India @ 75, 2021

⁸⁸ FICCI and EY-Parthenon, Transformation of Indian higher education, 2023; Brookings, Reviving Higher Education in India, 2019

⁸⁹ FICCI and EY-Parthenon, Transformation of Indian higher education, 2023; Brookings, Reviving Higher Education in India, 2019

 Accelerate multidisciplinary curricular offerings	Make teaching-learning more relevant, dynamic and interdisciplinary⁹⁰ - Strengthen the industry relevance of the curriculum envisaged by the National Higher Education Qualifications Framework including through industry 4.0-relevant courses (e.g., entrepreneurial development and AI) as well as complementary soft skills (e.g., cultural intelligence and inclusive leadership) and incorporating internships (for both professional, technical, and research-focused courses) for practical experience - Ensure such curricular offerings are updated through feedback from domain experts and inputs from industry, faculty, and students - Accelerate the uptake of online courses by students through credit transfer mechanisms and supporting more universities offering courses and micro-credentials through the SWAYAM platform	Integrate vocational education into mainstream HE⁹¹ - Encourage integration of VE concepts into curriculum of HE academic degrees, and introduce VE course offering by HEIs independently or through collaboration with industry and NGOs, as guided by the NEP - Improve linkages of vocational education with credits earned, and thereby enhance future employability of VE students in line with recent government efforts like the Academic Bank of Credits (ABC) and National Higher Education Qualifications Framework
 Increase access, equity, and affordability	Ensure equitable access to quality higher education among SEDGs⁹² - Scale-up non-collegiate institutions like polytechnics and vocational institutes, and enable wider uptake of cooperative degree and apprenticeship programs to enhance access - Enhance student loan financing options for SEDG students in both public and private HEIs through innovations in margin amounts and credit guarantees to banks and technology-based lending institutions (e.g., through the Central Sector Interest Subsidy Scheme (CSIS) and Credit	Enhance access and availability of online education - Provide students with the flexibility to undertake online courses through ed-tech platforms and expand the reach of Open and Distance Learning (ODL) to offer quality education beyond geographical limitations - Collaborate with ed-tech firms to build low-cost and widely applicable technology tools for integration into both academic and non-academic activities within HEIs, and facilitate access to devices through central and state scholarships

⁹⁰ NITI Aayog, Strategy for New India @ 75, 2021; FICCI and EY-Parthenon, Transformation of Indian higher education, 2023

⁹¹ Ministry of Education, About - Academic Bank of Credits (ABC), Accessed July 2023; University Grants Commission, Draft National Higher Education Qualifications Framework (NHEQF), 2022

⁹² World Bank, From Good to Great in Indian Tertiary Education: Realizing the Promise of the National Education Policy, 2023

	Guarantee Fund Scheme for Education Loan (CGFSEL)) • Support HEIs in creating more degree programs taught in Indian languages and bilingual formats • Establish guidelines for socio-emotional and academic support for students from disadvantaged backgrounds through counselling and mentoring programmes⁹³	• Enable students to build flexible online learning pathways through existing content libraries like SWAYAM
 Enhance faculty training and incentives	Adopt stringent faculty recruitment processes and norms⁹⁴ • Attract talented educators through reformed and rigorous selection processes such as by enhancing the efficiency of the UGC-recognised NET by reducing procedural barriers and focusing solely on assessing academic and research aptitude. • Enable the enrolment of international educators and subject experts for global and industry insights • Strengthen pre-service faculty training efforts to include exposure to contemporary pedagogical and research techniques, supplemented by continuous, flexible training and outcome-based faculty evaluation	Clarify academic career growth and incentives for greater faculty retention⁹⁵ • Support the development of a national framework for academic career growth that clarifies incentives such as security of tenure, clear advancement plans, and sufficient compensation for HEI teachers
 Advance state capacity for internationalisation efforts	Enhance inter-ministerial coordination and monitoring⁹⁶ • Enable coordination across major policy stakeholders like the UGC, the AICTE, and the Ministries of Education, Electronics and IT, Commerce and Industry, and External Affairs among others through working groups or task forces • Define and monitor performance benchmarks to evaluate the experience and progress on international universities set-up and student mobility	Develop an enabling regulatory, accreditation, and partnership ecosystem to attract international campuses⁹⁷ • Clarify and ease administrative processes (e.g., certificate issuance and work permits to attract), concession models for PPP investments, and mutual recognition of academic qualifications to increase new branch campuses and student exchange programs.

⁹³ Brookings, Unlocking young women's economic potential through digital mentoring in India, 2021

⁹⁴ <https://www.niti.gov.in/sites/default/files/2023-03/Strategy-for-NewIndia.pdf>

⁹⁵ Brookings, Reviving Higher Education in India, 2019; World Bank, From Good to Great in Indian Tertiary Education: Realizing the Promise of the National Education Policy, 2023

⁹⁶ World Bank, From Good to Great in Indian Tertiary Education: Realizing the Promise of the National Education Policy, 2023

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